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Ethical Hacking

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Ethical Hacking

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1. What is Hacking ?
2. Difference between Hacker & Cracker
3. What is Ethical Hacking ?
4. Hacker's Skill Set

What Is Hacking ?



WHAT DO YOU THINK, WHO IS A HACKER ?



OR



What Is Hacking ?



Getting into a system/network using unauthorised methods to gain access to sensitive information, *BUT...*



Hacker

Cracker

- Technically Skilled Professional.
- Scan for vulnerabilities & pitfalls in the system/network architecture, with permissions.
- Doesn't have any malicious intentions.
- Hacking is done for productive causes.

- Technically Skilled Professional.
- Scan for vulnerabilities & pitfalls in the system/network architecture, without permissions.
- Has any malicious intentions.
- Cracking is done for thefts/damaging systems/fun.

What Is *Ethical* Hacking ?



Ethical Hacker is a skilled professional who hacks into a system/network to scan for pitfalls and the probable targets that a **Cracker** might exploit.

Ethical Hacking $\approx$ Hacking

Ethical Hacking $\neq$ Cracking

IMPORTANCE OF ETHICAL HACKING

- Helps to maintain privacy & full control.
- Helps to find system flaws and thereby avoiding leaking sensitive information to Crackers.
- Since, computers are getting accessible to more and more people, ethical hacking helps to maintain and strengthen computer security.
- Get appointed at esteemed government as well as multi-national organisations.

Knowledge about:

- Network Protocols like HTTP, HTTPS
- Authentication Techniques
- Firewall Architecture, Port Details
- Network Architectures
- Web Server Configurations
- Web Application Structures
- Database Setups with Client-Server Architecture
- HTML, JavaScript, Python, Ruby
- Shell Scripting
- *And More...*

Ethical Hacking Tools

- Vulnerability Scanners and their benefits
- Types of Vulnerability Scanners
- Password Cracking Tools
- Packet Sniffers
- Popular Hacking Tools
- Hacking Hardware

What are Hacking Tools?



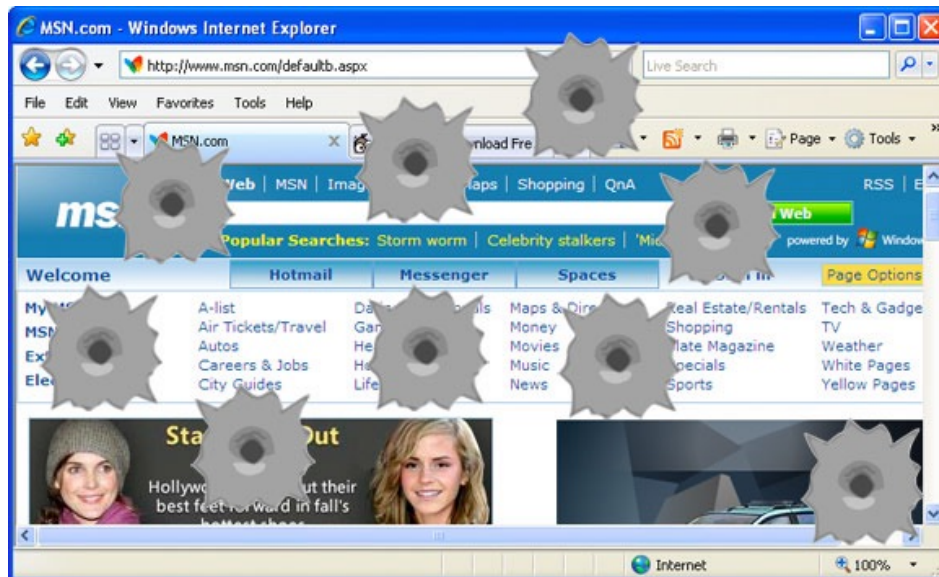
- ❖ Hacking tools are software applications designed to serve one or several specific purposes to hack/crack. These are used to make complex hacking procedures, easy-to-use and nowadays, also offer good GUI to help beginners in Ethical Hacking.

- ❖ There are numerous types of hacking tools like:
 - 1) Vulnerability Scanners
 - 2) Port Scanners
 - 3) Web Application Scanners
 - 4) Password Cracking Tools
 - 5) Packet Sniffers

Vulnerability Scanner



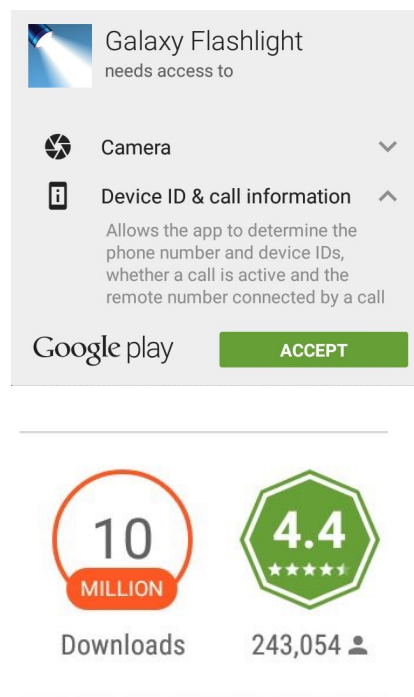
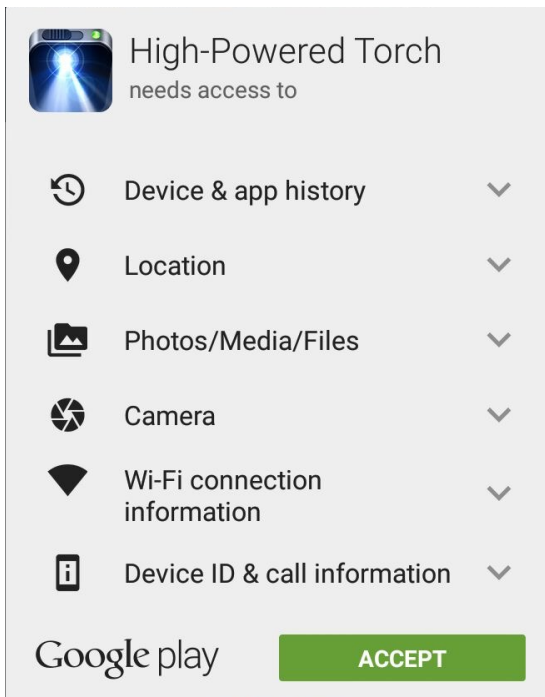
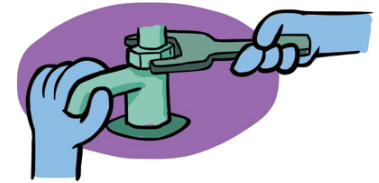
- ❖ Vulnerability can be defined as an unknown flaw in software, hardware or network which can be exploited to gain unauthorised access.
- ❖ Vulnerability Scanner is a computer program that detects weak spots in a network or computer system as a whole.
- ❖ Since, all the communication is done using ports on the network, the vulnerability scanners detect open ports which can be used to exploit weakness in the computer system.



Benefits of Vulnerability Scanners

Run a software and show its output here.

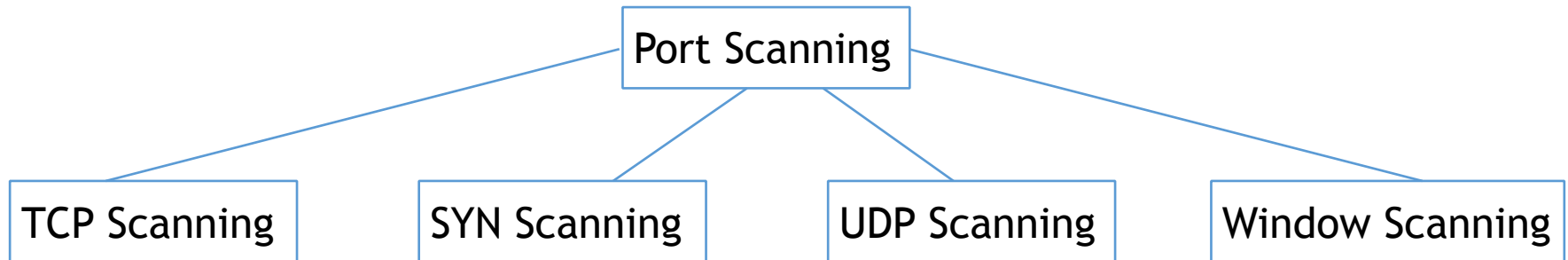
- ❖ Quick and Easy Detection of Weak Spots in computer system.
- ❖ Since, it helps in identifying the vulnerabilities, therefore, in-turn the vulnerabilities can be fixed.
- ❖ Keeping a frequent check on the computer system.
- ❖ Example: Top 10 flashlight apps on Google Play Store.



This was found by Gary Miliefsky at Snoop Wall with the help of Vulnerability Scanners. These apps install Trojans that cannot be removed without doing Factory Reset.

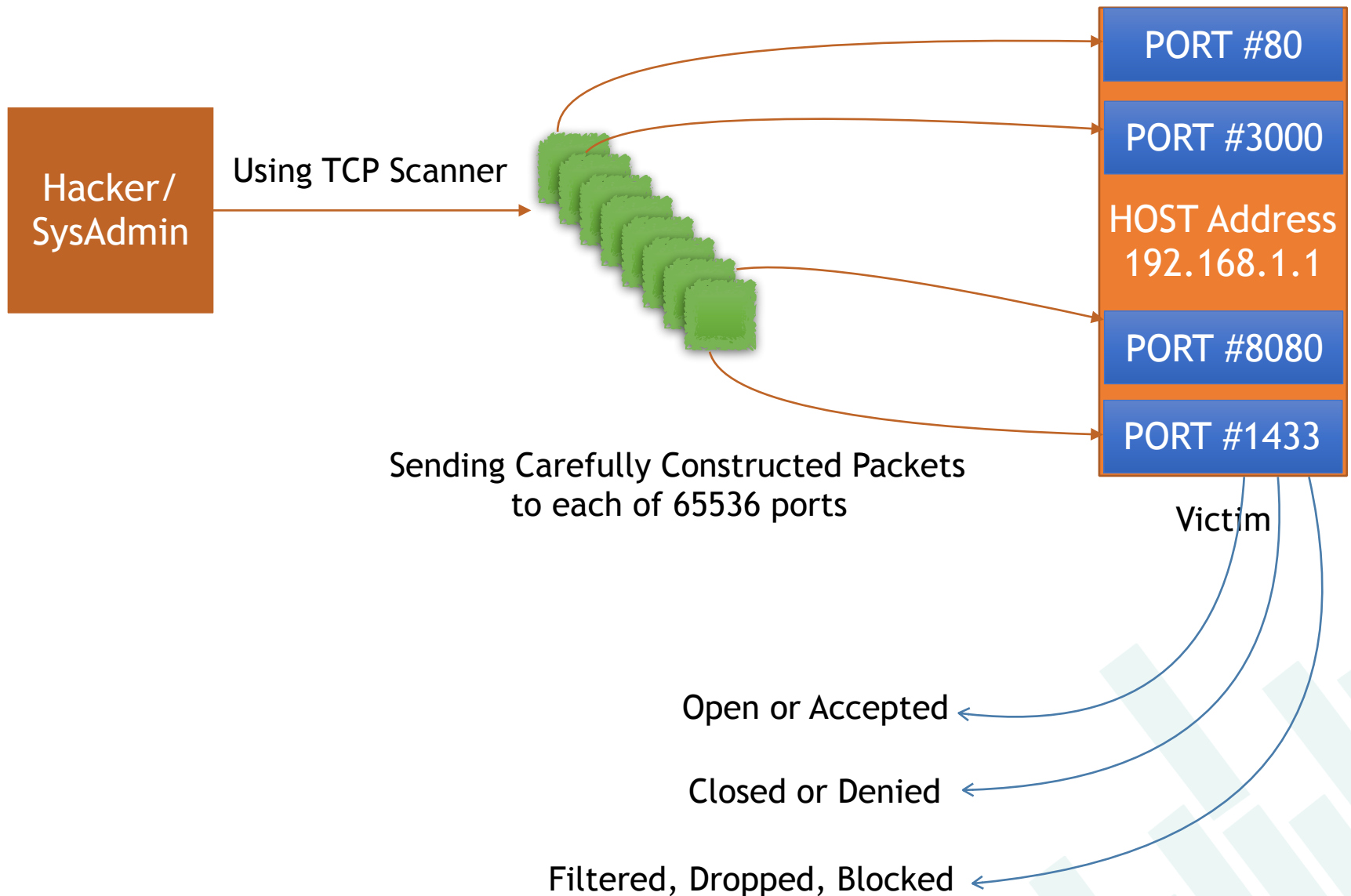
All information is sucked up by the servers mostly located in China, India and Russia.

Types of Vulnerability Scanners



- ❖ Port Scanner is a software program that investigates(not attack) a server for open and active ports by sending client requests to a range of server port addresses. It is used to verify security policies or compromise the services running on the ports.
 - ❖ Port Sweep is the technique of scanning multiple hosts for particular listening ports. Eg: SQL-based worms port sweep on multiple servers looking for TCP port number 1433.

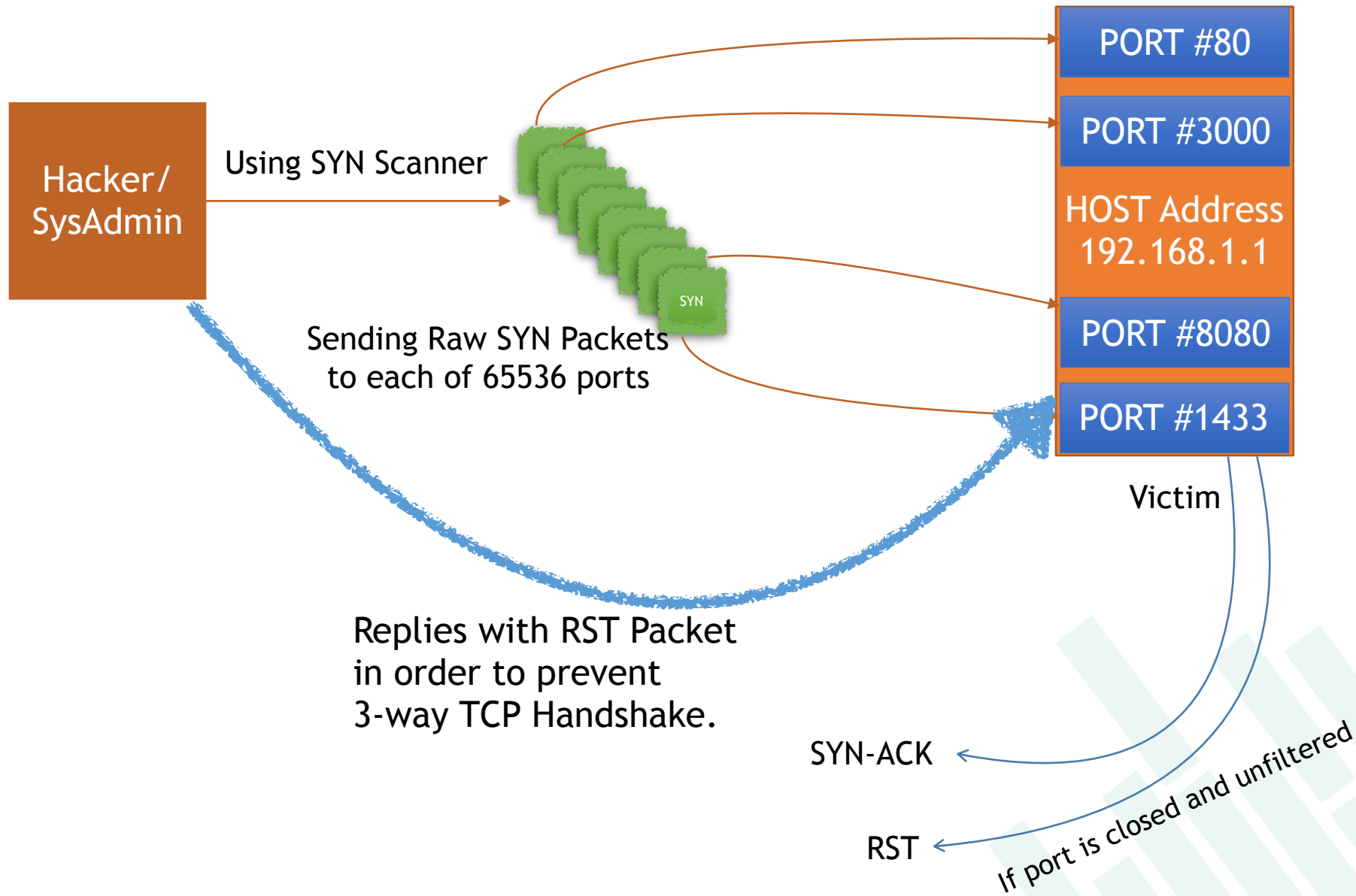
Types of Vulnerability Scanners: TCP Scanner



Types of Vulnerability Scanners: TCP Scanner

- ❖ TCP Scanner uses network functionalities of the operating system to carry out the desired procedure.
- ❖ Procedure: Whenever the port on the host machine is open, TCP scanner immediately closes the connection, if TCP three-way handshake is completed, in order to prevent performing DoS(Denial-of-service) attack. Because, if the connection is not closed an error code is generated and registered in logs which can be detected by the system administrator.
- ❖ Advantage: Special level user privileges are not necessary.
- ❖ Disadvantages:
 - Using OS functionalities, renders low-level control.
 - “Noisy” method i.e. the services log the attacker’s IP Address and IDS can raise alarm.

Types of Vulnerability Scanners: SYN Scanner



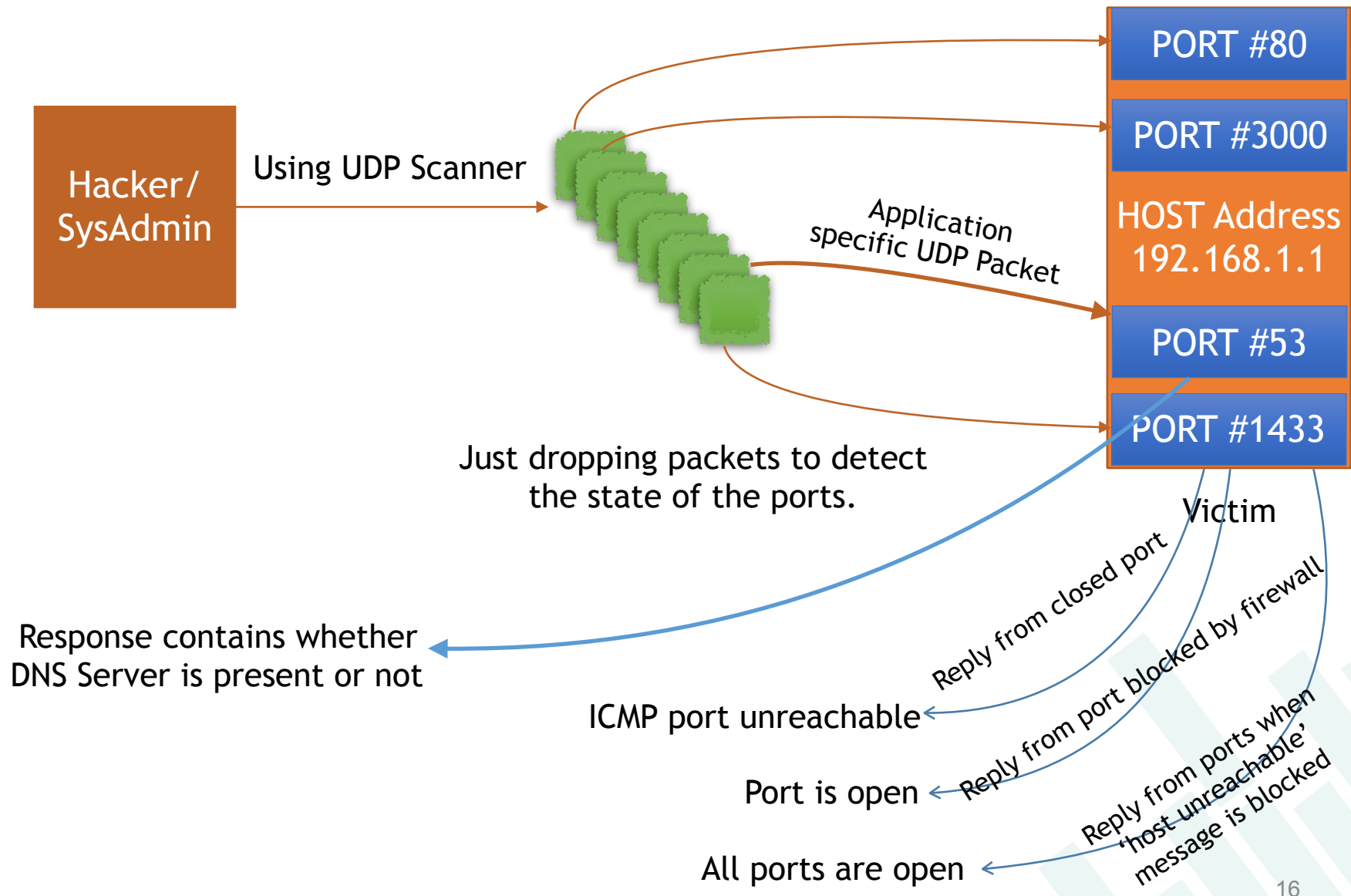
Types of Vulnerability Scanners: SYN Scanner

❖ Advantages:

- SYN Scanner has full control of the packets that are sent to the host and can also monitor them like response timeouts, response details, etc.
- Services running on the ports never actually establish a connection. SYN scanning is also called ‘Half-Open Scanning’ as full TCP connection is never established.
- RST packet can cause damage to the network devices like printers, scanners.

❖ Disadvantages: SYN Scanner need privileges to generate raw packets that initiate the whole process.

Types of Vulnerability Scanners: UDP Scanner



Types of Vulnerability Scanners: UDP Scanner

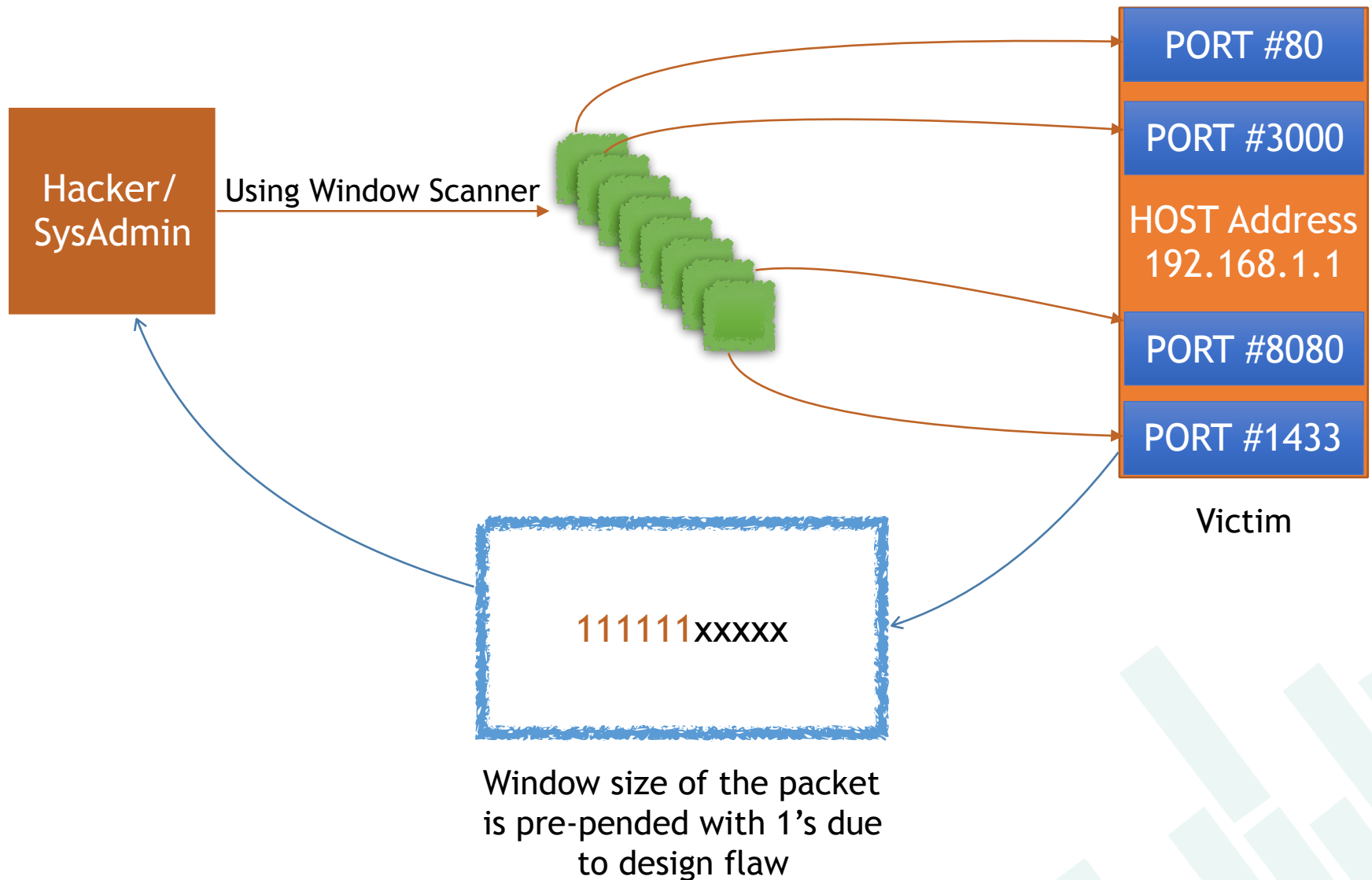
❖ Advantages:

- UDP Scanner can detect the number of open and closed ports by just dropping the packets and monitoring their responses.
- Connection-less protocol.

❖ Disadvantages:

- UDP application-specific scanning is limited by the number of ports for which application specific investigation packet is available or not.
- Technical Challenges are involved.

Types of Vulnerability Scanners: Window Scanner



Types of Vulnerability Scanners: Window Scanner

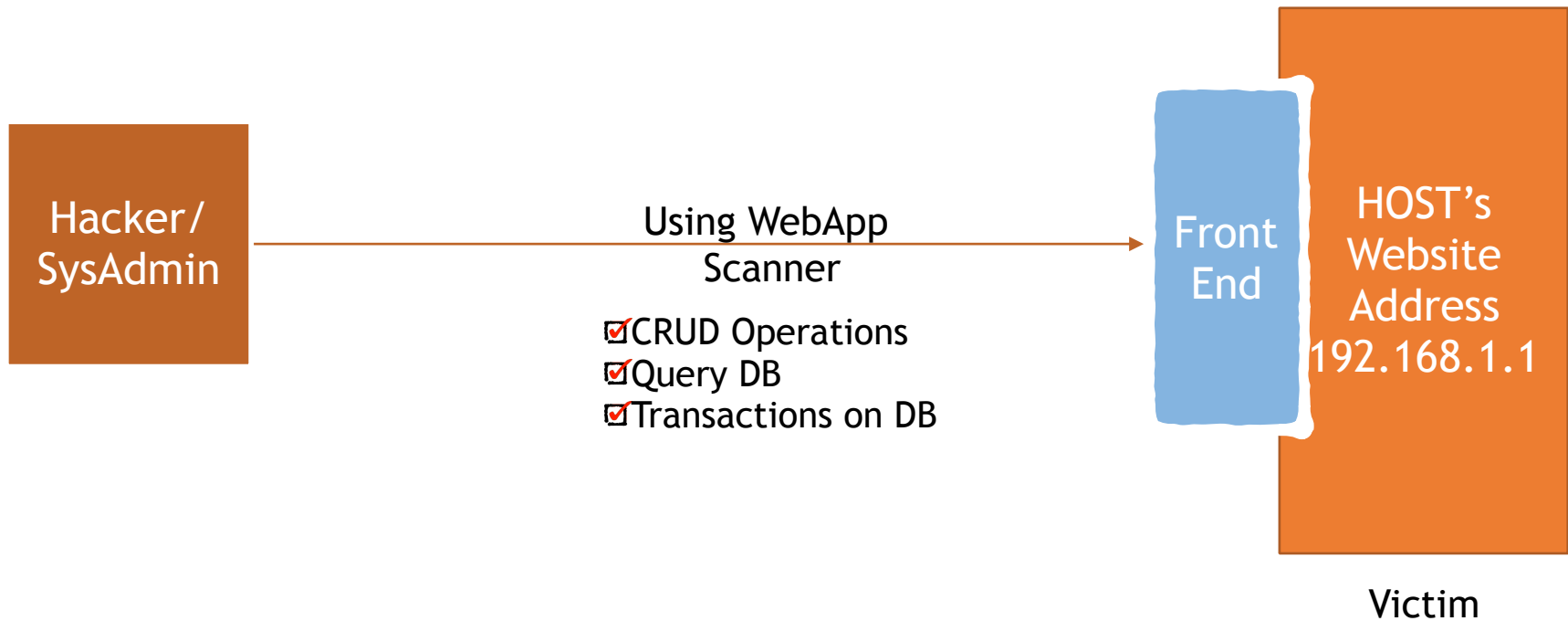
❖ Advantages:

❖ Disadvantages:

- Outdated
- Not trustworthy
- Nowadays, systems return the packet with pre-pended 0's in the window field, therefore, signalling all the ports as closed.



Types of Vulnerability Scanners: WebApp Scanner



- Enables **Black Box Testing**.
- Black Box Testing is a kind of application/ system testing where the tester has no clue about the business logic or processing of the web application/software and just have the information about the input and output.
- Scanner interacts with the UI and architecture only but doesn't scan source code.

Types of Vulnerability Scanners: WebApp Scanner

- ❖ Demo.
- ❖ Advantages:
 - ❖ Very handy tools which enable quick testing of the overall web application with less expertise.
 - ❖ Full stack, from FrontEnd to BackEnd, can be tested with ease.
 - ❖ Have good GUIs.
- ❖ Disadvantages:
 - ❖ The same tools are used by the attackers, thereby, bringing user and attacker on the same level.
 - ❖ These are automated tools and help attackers to quickly test/find vulnerabilities and send botnets.
 - ❖ Less efficient and reliable as these tools are not regularly updated at the same pace, with web application frameworks.
 - ❖ Cannot detect logical errors in the source code.